

COSC 39: Theory of Computation

Credit Statement: Talked to Sophie Park'27, Carson Bates'27, and Kiran Jones'27.

Problem 1. Let L be an arbitrary automatic language. Prove that the following languages are also automatic.

1. $\text{nosuffix}(L) := \{w \in L : \text{no proper suffix of } w \text{ is in } L\}$
2. $\text{suffix}(L) := \{y \in \Sigma^* : xy \in L \text{ for some } x \in \Sigma^*\}$
3. $\text{dropout}(L) := \{xz \in \Sigma^* : xyz \in L \text{ for some } x, z \in \Sigma^* \text{ and } y \in \Sigma\}$

Solution.

1. Let D be a DFA that recognize the automatic language L , described by the tuple $(Q, s, A, \Sigma, \delta)$. We now construct a new ε -DFA N that recognizes the language $\text{rev}(\text{nosuffix}(L))$ as follows:

- $Q_N = Q_D \cup \{s_N\}$.
- We add a new start state called s_N with ε -transitions to all $a \in A_D$ where A_D were the accepting states in D .
- $A_N = \{s_D\}$, where s_D is the start state for D .
- For every transition $\delta_D(q, a) = q'$ in D we add a transition $\delta_N(q', a) = q$ in N . And we remove every transition from p to other states q (even if $p = q$ if p is an accepting state in D . That is,

$$\delta(p, a) = \emptyset$$

for every $p \in A$ and $a \in \Sigma$.

We argue that N accepts a string w if and only if $\text{rev}(w) \in L$ and has no proper suffixes in L .

($\implies$)

Assume N accepts w . Then there exists a path $\delta_N^*(s_N, w) = s_D$ where,

$$s_N \xrightarrow{\varepsilon} q \xrightarrow{w} s_D$$

for some $q \in A_D$. Then by construction (and reversal of transitions) we have $\delta^*(s_D, \text{rev}(w)) = q \in A_D$. So $\text{rev}(w) \in L$.

For contradiction, assume that $\text{rev}(\mathbf{w})$ has a proper suffix $\text{rev}(\mathbf{y})$ such that $\text{rev}(\mathbf{y}) \in L$. Then we can write

$$\text{rev}(\mathbf{w}) = \text{rev}(\mathbf{x}) \text{rev}(\mathbf{y}),$$

so $\mathbf{w} = \mathbf{yx}$, where $\mathbf{y} \neq \mathbf{w}$. Then in D we have,

$$s_D \xrightarrow{x} q \xrightarrow{y} q' \in A_D.$$

But the reverse of this path in N gives us a path

$$A_D \ni q' \xrightarrow{\text{rev}(\mathbf{y})} q \xrightarrow{\text{rev}(\mathbf{x})} s_D \in A_N$$

Since $q' \in A_D$, there is an ε -transition from s_N to q' . So in N , we can start at q' and follow the path $\text{rev}(\mathbf{y})$ to reach q .

But by construction, we removed all outgoing transitions from accepting states in D . Thus, for any $\mathbf{a} \in \Sigma$, we have

$$\delta_N(q', \mathbf{a}) = \emptyset.$$

This means that $\text{rev}(\mathbf{w}) \notin L(N)$, a contradiction. Hence, $\text{rev}(\mathbf{w})$ has no proper suffix in L , and hence $\text{rev}(\mathbf{w}) \in \text{nosuffix}(L)$.

($\Leftarrow$)

Assume $\mathbf{w} \in \text{nosuffix}(L)$. Then there exists some path $\delta^*(s_D, \mathbf{w}) = q \in A$, and no proper suffix of $\mathbf{w}$ is in L , meaning there is no shorter subset of this path that leads to acceptance. Then by construction $\delta'_N(s_N, \text{rev}(\mathbf{w})) = s_D \in A_N$. Hence, $\text{rev}(\mathbf{w}) \in L(N)$.

Hence, $L(N) = \text{rev}(\text{nosuffix}(L))$ and it is automatic. Moreover from the previous worksheet we know that $\text{rev}(\text{rev}(\text{nosuffix}(L))) = \text{nosuffix}(L)$ is also automatic.

2. Let D be a DFA that recognize the automatic language L , described by the tuple $(Q, s, A, \Sigma, \delta)$. We now construct a new NFA $D' = (Q', S', A', \Sigma', \delta')$ that recognizes the language $\text{suffix}(L)$ as follows:

- States $Q' : Q$
- Starting states $S' : Q$ (every state is a start state)
- Accepting states $A' : A$
- Alphabet $\Sigma' : \Sigma$
- Transition function $\delta' : \delta$

We argue that D' accepts a string if and only if it is a suffix of a string D accepts.

($\implies$)

Assume D' accepts w . Then $\delta^*(q, w) \in A$ for some state q . But by construction of states and transitions we know that q is a state of the DFA D and there exists a string $x \in \Sigma^*$ such that $\delta^*(s, x) = q$. Then,

$$\delta^*(s, xy) = \delta^*(q, y) \in A$$

So $xy \in L$ and $y \in \text{suffix}(L)$.

($\Leftarrow$)

Assume D accepts xy , then $y \in \text{suffix}(L)$. Then $\delta^*(s, xy) \in A$. Let $q = \delta^*(s, x)$. Then $\delta^*(q, y) \in A$. By construction we know that q is a starting state in D' , and there exists a path $\delta'^*(q, y) \in A'$.

Hence, $y \in L(D')$ and $L(D') = \text{suffix}(L)$.

Note since we know that for every NFA there exists an equivalent DFA, $\text{suffix}(L)$ is automatic.

3. Let D be a DFA that recognize the automatic language L , described by the tupe $(Q, s, A, \Sigma, \delta)$. We now construct an NFA N that recognizes the languag $\text{dropout}(L)$ as follows:

- $Q_N = Q \times \{0, 1\}$.
- Start state: $(s, 0)$.
- Accepting states: $A_N = A \times \{1\}$.
- Transitions:
 - For all $q \in Q$ and $a \in \Sigma$,

$$\delta_N((q, 0), a) = (\delta(q, a), 0).$$

- For all $q \in Q$ and $a \in \Sigma$,

$$\delta_N((q, 1), a) = (\delta(q, a), 1).$$

- For all $q \in Q$, $a \in \Sigma$, and all $q'' \in Q$ such that there exists $q' \in Q$ and $b \in \Sigma$ with $\delta(q, a) = q'$ and $\delta(q', b) = q''$ we add a transition

$$\delta_N((q, 0), a) = (q'', 1).$$

The intuition is that we have two modes ‘0’ and ‘1’ where we are in mode 0 before skipping y and we are in mode 1 after skipping y (also note that there is no way to go from mode 1 to mode 0). We argue that N accepts w if and only if there exist $x, z \in \Sigma^*$ and $y \in \Sigma$ such that $xyz \in L$ and $w = xz$.

($\implies$)

Assume N accepts w . Then there exists some path $\delta_N^*((s, 0), w) \in A_N$. By construction, an accepting run must use exactly one transition of the form

$$(q, 0) \rightarrow (q'', 1),$$

since this is the only way to move from mode 0 to mode 1, and all accepting states lie in mode 1.

Hence w must be some string xyz where x is the part of the input read before the switch, y is the symbol skipped, and z is the part of the input read after the switch.

Since the transition $(q, 0) \rightarrow (q'', 1)$ is only defined when there exists a path in D of the form

$$\dots \rightarrow q \xrightarrow{a} q' \xrightarrow{y} q'' \xrightarrow{b} \dots$$

we have $xyz \in L$, and therefore $w = xz \in \text{dropout}(L)$.

($\impliedby$)

Assume $w = xz$ and there exists $y \in \Sigma$ such that $xyz \in L$. Then D has a path

$$s \xrightarrow{x} q \xrightarrow{y} q' \xrightarrow{z} a$$

for some $a \in A$. By construction, in N we have $\delta_N^*((s, 0), x) = (q, 0)$, and $\delta_N^*((q, 0), z) = (a, 1)$. That is, we read x in mode 0, take the transition that skips y and moves to mode 1, and read z in mode 1. Thus $w \in A_N$.

Hence $\text{dropout}(L) = L(N)$ and we know that for every NFA there exists an equivalent DFA and hence, $\text{dropout}(L)$ is automatic.